Muhammad Hassan

Drexel Hill, PA 19026 | (484) 474-7700 | msh345@drexel.edu | linkedin.com/muhammad-hassan | Portfolio

EDUCATION

Drexel University | Philadelphia, PA GPA: 3.70
Bachelor of Science in Mechanical Engineering | *Mathematics Minor* Anticipated Graduation: June, 2029
Honors & Awards: NSF AIME Scholar, Drexel Founders Scholar, Fannie M. & James Dewey Hardy Scholarship, Charles Husten Grant, William Utzy Scholarship
Relevant Coursework: CAD (SolidWorks), Mechatronics, Statics, Dynamics, Engr Design, Material Science, Thermodynamics, Calculus I, II, Multivariate Calculus, Differential Equations, Mechanics of Materials, Thermodynamic Analysis, Discrete Mathematics, Fundamentals of Physics I, II, III

TECHNICAL SKILLS

- **CAD & Simulation:** SolidWorks (3D Modeling, Assembly Mating, Exploded Views, Technical Drawings), AutoCAD, FEA Stress/Vibration Analysis, FMEA
- **Embedded Systems & Electronics:** Microcontrollers (Raspberry Pi Pico / Pico 2W, Arduino Mega), I2C Protocol, MOSFET Drivers, L298N H-Bridges, Sensor Integration (AHT 10, HC-SR04, OLED Displays)
- **Programming & Software:** Python, MATLAB, Excel (Advanced Formulas, Pivot Tables, Data Analysis)
- **Fabrication & Lab Work:** Prototyping, 3D Printing, Benchtop Electronics Testing, Hand & Power Tools, Quality Assurance / SOP Enforcement, Soldering

PROFESSIONAL EXPERIENCE

Exelon | Eddystone, PA | *Engineering Co-op – Construction & Maintenance* **March 2026 – September 2026**

- Developed custom Excel operational tools and work-order tracking dashboards, optimizing data tracking, eliminating workflow redundancies, and generating over $100k+ in estimated annual savings.
- Performed field engineering inspections and observational support alongside utility crews, evaluating power distribution equipment, transformer dimensioning, reclosers, and triple-phase primary lines.
- Cross-functionally supported understaffed operations across separate departments while maintaining original deliverables.

HERA | Philadelphia, PA | *HASP Payload Design / Drone Recovery Team Lead* **February 2026 – Present**

- Designed and engineered a high-altitude 20kg payload in SolidWorks optimized for structural resistance against multidirectional loading exceeding ≥10 Gs. thermal factors, and vibration profiles.
- Performed structural trade studies comparing tungsten and aluminum support structures to balance high-G mechanical stability, mass constraints, and thermal endurance.
- Co-authored the project technical abstract and presented findings for NASA Pennsylvania Space Grant Consortium documentation.

Cashier & Manager - Norwood One Stop - Norwood, PA **February 2021 – September 2025**

- Led independent oversight for the entire store, ensuring efficient process flow and maintaining high service standards during shifts.
- Managed comprehensive inventory control for 300+ SKUs, optimizing stock and minimizing product waste
- Displayed comprehensive financial accountability, monitoring all cash transactions and securing daily opening/closing procedures as defined in standard operating procedures.

PROJECTS

Infant Medicine Delivery System - Freshman Design Project

- Conducted comprehensive analysis and concept generation for a new infant drug delivery system, successfully incorporating a medicine reservoir and dispensing mechanism within a standard pacifier.
- Performed prototyping and fabrication, physically assembling multiple working models to test safety, fluid dynamics, and ergonomic constraints.